MIL-M-45531B(EL)
26 December 1968
Superseding
MIL-M-45531A(MO)
14 May 1964

MILITARY SPECIFICATION
METASCOPE, AN/PAS-6

1. SCOPE

1.1 This specification covers a night vision sight termed metascope.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636	- Box, Fiberboard

Military

MIL-E-1/1583	- Electron Tubes, Infrared Image Converter
MIL-B-18	- Batteries, Dry
MIL-P-116	- Preservation, Methods of
MIL-T-152	- Treatment, Moisture-and-Fungus-Resistant, of Communications, Electronics, and Associated Electrical Equipment
MIL-T-704	- Treatment and Painting of Material
MIL-P-45535	- Power Supply, Transistorized, 12 Kolovolt
MIL-E-55301	- Electromagnetic Compatibility

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Property

FSC 5855

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MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts

DRAWINGS

ECOM

SC-DL-635250 - Metascope, AN/PAS-6

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

AMERICAN TRUCKING ASSOCIATION
National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, 1616 P Street, NW, Washington, D.C. 20036.)

NATIONAL BUREAU OF STANDARDS

(Application for copies should be addressed to National Bureau of Standards, Springfield, Virginia 22151.)

3. REQUIREMENTS

3.1 Description. - The hand-held battery operated near infrared image forming receiver and infrared light source and a carrying case for both shall be as specified herein (see SC-DL-635250).

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3.1.1 Drawings. - The drawings forming a part of this specification are engineering design drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper fits, the contractor shall provide tolerances within those prescribed on the drawings to insure proper fit, assembly, and operation of the metascope. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.2 Preproduction Metascope. - The contractor shall furnish three metascopes, within the time frame specified in the contract, to prove prior to starting production, that his production methods and choice of design detail will produce metascopes that comply with the requirements of this specification. Any changes or deviation of metascopes from the preproduction metascope during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the contractor of his obligation to furnish metascopes conforming to this specification.

3.3. Material. - Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Temperature and Humidity. - The metascope shall operate and shall not be damaged (see 6.4.2) by operation and storage under the conditions set forth below:

Item	Temperature	Humidity
Operation	1. Temperature shocks from 125°F. (see 6.4.1) to room temperature (room temperature shall be defined as 70°F) and from minus 25°F to room temperature.	Maintain highest relative humidities possible at all temperatures above 70°F. Allow relative humidity to be uncontrolled for all temperatures below 70°F.
	2. Ambient temperatures from 115°F to minus 25°F. (A voltage from an external source may be supplied to the high-voltage power pack and light source at temperatures below plus 20°F.)	

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Item	Temperature	Humidity
Storage	Ambient temperatures from 155°F to minus 65°F.	Maintain highest relative humidities possible at these temperatures.

3.5 Immersion. - The metascope with batteries installed shall operate and shall show no evidence of the entrance of moisture when immersed to a depth of 3 feet in fresh water for a period of 30 minutes.

3.6 Resistance to Vibration. - The metascope shall not be damaged by vibration at a frequency of 30 cycles per second (cps) at an amplitude of not less than 0.0625 inch (total displacement of 0.1250 inch) for periods of 10 minutes each along each of the following axis:

a. Parallel to the optical axis.
b. Perpendicular to the optical axis in the vertical plane.

3.7 Infrared Receiver

3.7.1 Performance. - Performance of the receiver at 70°F with the high-voltage power supply applying no more than 12,000 volts, direct current (dc), to the image tube shall be as listed below:

3.7.1.1 Contrast Transfer Objective Lens: The contrast transfer shall be equal to or greater than the values listed below:

Test Points	LP/MM @ Photocathode	CT Values in Percent
1	2.5	96%
2	7.5	92%
3	16.0	88%
4	24.0	80%
5	32.0	72%

3.7.1.2 Contrast Transfer Eyepiece Lens. - The contrast transfer shall be equal to or greater than the values listed below:

Test Points	LP/MM @ Photocathode	CT Values in Percent
1	2.5	94%
2	7.5	82%
3	16.0	58%
4	24.0	40%
5	32.0	28%

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Linear distortion	- Not in excess of 8 percent at 12 degrees from center of field of view.
3.7.1.3 Field of view	- Not less than 26 degrees.
3.7.1.4 Focus range	- From 12 inches or less to infinity.

3.7.2 High-Voltage Power Supply. - The power supply shall be in accordance with MIL-P-45535.

3.7.3 Image Tube. - The image tube shall be type 6929 in accordance with MIL-E-1/1583.

3.7.4 Tube-Guide Insert. - The tube-guide insert, with the field corrector lens cemented in place, shall withstand 15,000 volts between the inner and outer surfaces without voltage breakdown occurring.

3.8 Infrared Light-Source

3.8.1 Beam Characteristics. - With a test voltage of 2.7 volts, dc, applied to the lamp 10 feet from a flat surface, the light-beam characteristics of the lamp and reflector assembly minus the infrared filter shall meet the following requirements.

Peak-beam candlepower	- Not less than 1,000.
Beam candlepower distribution	- Not less than 1 percent of peak candlepower throughout the base of a cone of the beam having a 16-degree vertex angle.
Hot-spot concentricity	- Not more than 2.86 degrees of the outer ring of light.

3.8.2 Infrared Filter. - The infrared filter shall have a transmittance of not less than 60 percent at 1.03 microns and not more than 5 percent at 0.8 micron and cut off at 0.72 micron.

3.8.3 Alinement. - The infrared light-source when attached to the receiver shall produce a light-beam parallel to within 1.0 degree to the optical axis of the receiver.

3.9 Batteries. - Batteries shall be in accordance with MIL-B-18. The type BA-30 battery shall be used for the infrared light-source and the type BA-1312/U battery shall be used for the power supply. Unless otherwise specified (see 6.2), the batteries shall be furnished by the contractor.

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3.10 Electromagnetic Compatibility. - The electromagnetic compatibility of the metascope shall be in accordance with MIL-E-55301, Tactical Support Class.

3.11 Fungus Resistance. - The metascope, except for the batteries, canvas bag, electrical components, such as switches and contacts, lamps, reflecting surfaces, radiation transmitting elements, and components hermetically sealed, shall be treated for fungus resistance in accordance with MIL-T-152.

3.12 Identification Markings. - Identification markings shall be in accordance with MIL-STD-130.

3.13 Technical Publications. - Such technical publications as are specified shall be furnished (see 6.2).

3.14 Repair Parts. - Such repair parts as are specified shall be furnished (see 6.2).

3.15 Items to be Furnished with the Metascope. - No shipment of metascopes shall be made unless such repair parts, publications, and accessories which are to be furnished with each unit as specified in the contract or order are included with the shipment or unless approval for shipment of the metascope without such items has been received from the contracting officer.

3.16 Treatment and Painting. - The portions of the metascope and its components and parts normally painted shall be cleaned, treated, and painted in accordance with the applicable part drawing and in accordance with MIL-T-704.

3.17 Workmanship. - Joints and seams shall be tight fit; edges shall be smoothly finished. The interior of the housings shall be free of all foreign matter. The optical surfaces shall be free of all scratches, striae, mold marks, or blemishes which may affect their optical qualities, mechanical strength, or good appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed

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necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. - Inspection shall be classified as follows:

(a) Preproduction Inspection (see 4.3)
(b) Quality Conformance Inspection (see 4.4.)
(c) Inspection Comparison (see 4.6)
(d) Inspection of Preparation for Delivery (see 4.7)

4.3 Preproduction Inspection.

4.3.1 Examination. - Each Metascope shall be examined for the following defects. One or more defects shall be cause for rejection of that unit.

MAJOR

101. Objective binds while focusing.
102. Eyepiece diopter adjustment binds.
103. External hardware loose.
104. Viewer does not operate when turned on.
105. Light source does not operate when turned on.

MINOR

201. Finish on viewer and/or light source and carrying case does not conform to MIL-STD-171 and applicable drawings.
202. Identifications and marking do not conform to MIL-STD-130 and applicable drawings.

4.3.2 Tests. - Upon successful completion of the examination specified in 4.3.1, the three preproduction models shall be subjected to the tests of Table I.

4.3.3 Inspection Failure. - Failure of the preproduction model to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.3.1 and 4.3.2, respectively, shall be cause for rejection.

4.4 Quality Conformance Inspection

4.4.1 Sampling. - Sampling for examination and tests shall be in accordance with MIL-STD-105. A lot shall be formulated from metascopes which have completed the individual tests specified in 4.4.3.1. Inspection level S-2

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shall be used for lot sizes 2 to 25; inspection level S-3 shall be used for lot sizes 26 or larger. AQL for examination and test together shall be 6.5 percent defective.

TABLE I PREPRODUCTION TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Vibration	4.5.1.1	3.6
Immersion	4.5.1.4	3.5
Distortion	4.5.2.3	3.7.1
Field of View	4.5.2.4	3.7.1
Focus Range	4.5.2.5	3.7.1
Tube-Guide Insert	4.5.3	3.7.4
Beam Characteristics	4.5.2.6	3.8.1
High Temperature & Humidity	4.5.1.2	3.4
Low Temperature & Temperature Shock	4.5.1.3	3.4
Contrast Transfer Objective	4.5.2.1	3.7.1
Contrast Transfer Eyepiece	4.5.2.2	3.7.1
Infrared Light Source Alinement	4.5.4	3.8.3
Infrared Filter	4.5.5	3.8.2
Electromagnetic Compatibility	4.5.6	3.10

4.4.2 Examination. - Samples selected in accordance with 4.4.1 shall be examined in accordance with 4.3.1 and compared with the preproduction model. Each sample which deviates from the preproduction models shall be considered a defective sample.

4.4.3 Tests.

4.4.3.1 Group A. - Each metascope shall be tested in accordance with Table II. Failure of any test shall be cause for rejection.

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TABLE II GROUP A TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Tube-Guide Insert	4.5.3	3.7.4
Contrast Transfer Objective	4.5.2.1	3.7.1
Contrast Transfer Eyepiece	4.5.2.2	3.7.1

4.4.3.2 Group B. - Group B tests shall be conducted on metascopes selected from units which have passed the examination and tests in 4.4.2 and 4.4.3.1. Samples shall be selected in accordance with 4.4.1. Tests shall be conducted in accordance with Table III. Failure of any test shall be cause for rejection.

TABLE III GROUP B TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Vibration	4.5.1.1	3.6
Immersion	4.5.1.4	3.5
Contrast Transfer Objective	4.5.2.1	3.7.1
Contrast Transfer Eyepiece	4.5.2.2	3.7.1

4.4.3.3 Group C. - This inspection shall consist of the test specified in Table IV and shall be performed on sample units that have been subjected to and met Group A and B inspection. Samples shall be selected in accordance with 4.4.1.

TABLE IV GROUP C TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
High Temperature & Humidity	4.5.1.2	3.4
Low Temperature and Temperature Shock	4.5.1.3	3.4
Tube-Guide Insert	4.5.3	3.7.4
Field of View	4.5.2.4	3.7.1
Focus Range	4.5.2.5	3.7.1

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TABLE IV (Cont.)

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Beam Characteristics	4.5.2.6	3.8.1
Infrared Light Source Alinement	4.5.4	3.8.3
Infrared Filter	4.5.5	3.8.2
Contrast Transfer Objective	4.5.2.1	3.7.1
Contrast Transfer Eyepiece	4.5.2.2	3.7.1

4.5 Test Procedures

4.5.1 Environmental.

4.5.1.1 Vibration. - Mount the metascope rigidly to the vibration table. Subject the metascope to a simple harmonic motion parallel to the optical axis and perpendicular to the optical axis in the vertical plane at an amplitude of 0.0625 inch (0.1250 inch total excursion) at frequency of 30 cycles per second (cps) for 10 minutes in each position. Examine the metascope. Evidence of damage shall constitute failure of this test.

4.5.1.2 High Temperature and Humidity. - The metascope shall be subjected to the temperature and humidity cycling test specified in MIL-STD-202, Method 106A, with the following modifications:

(a) Eliminate the initial conditioning in the dry oven, and steps 7a and 7b.
(b) Run steps 1 through 6 with exception of the last cycle through all temperatures from room temperature to 155°F.
(c) During the last cycle, reduce the maximum temperature to 115°F, install all batteries, and allow the equipment to operate throughout all temperatures from room temperature to 115°F.
(d) Raise temperature to 125°F and remove to room temperature.

The metascope shall be examined for damage. Any evidence of damage shall constitute failure of this test.

4.5.1.3 Low Temperature and Temperature Shock. - The metascope shall be placed in a test chamber and the internal temperature of the chamber lowered to minus 65°F in not less than 2 hours. During the period of

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temperature change from room temperature to minus 25°F, the metascope shall be turned on and visually inspected for functioning of the image tube and easy movement of switches at plus 25°F and again at minus 25°F. After 72 hours, the temperature shall be raised to minus 25°F and held for at least 2 hours, at which time the metascope shall be removed to room temperature. The metascope shall be examined for damage at the conclusion of the test. Any evidence of damage shall constitute failure of this test.

4.5.1.4 Immersion. - The infrared receiver with battery and the infrared light source with its batteries shall be immersed to a depth of 3 feet in fresh water for a period of 30 minutes. After removal from the water, the light source and receiver shall be examined for entrance of moisture and operation. Any evidence of the entrance of moisture or failure to operate shall constitute failure of this test.

4.5.2 Infrared Receiver. - The following tests shall be made in an ambient temperature of 70°F plus or minus 5°F and with a potential of not more than 12,000 volts applied to the image tube by the power supply.

4.5.2.1 Contrast Transfer-Objective Lens. - Place the viewer between the collimator and eye-corrected photo-detector as shown in Figure 1. Illuminate the frequency pattern with light from a 2870°K tungsten lamp and focus the image of the frequency pattern at the slit, reducing the aperture of the lens used with the phototube to a diameter of 7 mm. Adjust the X-Y recorder for zero level (see Figure 2) by blocking the entrance of all light to the photodetector. Advance the frequency pattern (see Figure 3) through the five frequencies specified in 3.7.1 at a constant rate, such that no signal damping due to response time of the image converter assembly occurs, and obtain a recording similar to that shown in Figure 2. Calculate the contrast transfer using the formula $\frac{W-B}{W+B} \times 100$ from the measured data.

(NOTE: Special care must be taken before test data is made to avoid losses resulting from the test instrument.) At the test points specified complete and record the contrast transfer values. Failure to comply with values specified in 3.7.1 shall constitute failure of this test.

4.5.2.2 Contrast Transfer-Eyepiece. - Perform the eyepiece lens contrast transfer test as described in 4.5.2.1. Failure to comply with the values specified in 3.7.1 shall constitute failure of this test.

4.5.2.3 Distortion. - The following procedure shall be used to measure distortion:

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(a) Mount the receiver in a rigid position as shown in Figure 4 so that it is on the optical center with the collimator, telescope, and target pattern shown in Figure 5.

(b) With the telescope, measure angular height at center of target.

(c) Rotate the telescope around entrance pupil to a line 12 degrees from center.

(d) With telescope, measure the angular height of line 12 degrees from center.

(e) Compute distortion by the formula:

$$\text{Percent distortion} = \frac{h_1 - h_2 \times 100}{h_1}$$

Where: h_1 = Height of centerline

h_2 = Height of line 12 degrees from centerline

Linear distortion in excess of 8 percent at 12 degrees from the center of the field of view shall constitute failure of this test.

4.5.2.4 Field of View. - The field of view shall be examined by observing with the infrared receiver two vertical lines placed 55.4 inches apart on a vertical surface. The test is to be conducted with the target lines illuminated with visible radiation (white light). The lines shall be 3/4 inch wide and shall be black on a white background. When both lines are seen at a distance of 10 feet with the viewer held rigid in one position, the field of view is acceptable. A field of view less than 26 degrees shall constitute failure of this test.

4.5.2.5 Focus Range. - With the objective-lens assembly of the receiver adjusted to the full counterclockwise position, the distance from the objective lens to a focused object shall be measured. This distance shall be known as the minimum-focus range. The objective-lens assembly shall then be turned slowly to the full-clockwise position while the receiver is pointed toward an object at a distance of not less than 100 yards. When a well-focused image of the distant object is obtained before the full-clockwise position is reached, and the aforementioned minimum-focus range is equal to or less than 12 inches, the focus range is acceptable.

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Failure to obtain a focus range from 12 inches or less to infinity shall constitute failure of this test.

4.5.2.6 Beam Characteristics. - The metascope light-source, energized with 2.7 volts, dc, shall be set in a stationary position at a distance of 10 feet from a vertical wall such that the wall is illuminated with the beam perpendicular to it. Using a standard photometer (see 6.3.1), the peak foot-candles (or maximum hotspot) shall be measured by moving the meter probe throughout the light-beam at the wall surface. Any photometer used shall be calibrated against a National Bureau of Standards (NBS) standard lamp with a color temperature of 2870°K before tests are made. The formula for converting foot-candles to candlepower is:

$$I = E\,d^2$$

I = Candlepower of the lightsource

E - Foot-candles measured by the photometer

d = Distance from the light source to the photometer

Peak-beam candlepower less than 1,000 shall constitute failure of this test. Find the same maximum hotspot using the Densichron S4 probe (see 6.3.1) and adjust the linear percent scale of the Densichron to indicate 100 percent. The probe is to be used without the aperture attachment. Measure and record the minimum percent reading found by moving the probe throughout a 34-inch-diameter circle concentric with the hotspot. Use the decade multiplier switch to keep the minimum reading within a readable range for accuracy. A minimum reading less than 1 percent of the hotspot shall constitute failure of this test. An outer ring of light cast onto the wall not within 6 inches of concentricity of the hotspot shall constitute failure of this test.

4.5.3 Tube-Guide Insert. - With the image converter tube and objective lens assembly removed from the viewer housing, a negative 15,000 volts shall be applied to a conductor placed in contact with the high-voltage alinement ring and also in contact with the field corrector lens seam. A ground potential probe shall then be used to explore the outer surface of the housing, in the vicinity of the high voltage alinement ring, and the tube-guide insert at the objective end, with particular care taken to test the field corrector lens seam. Sparking or other evidence of electrical breakdown shall constitute failure of this test.

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4.5.4 Infrared Light-Source Alinement. - With the infrared light-source attached to the receiver, the metascope shall be held in a rigid position so that the infrared-beam is projected on a vertical surface 100 inches from the front of the receiver. The center of the field of view shall be marked on this surface and a circle of 4 inches in radius centered at this point. The apparent center of the infrared-beam as seen through the receiver shall then fall within this circle for the alinement to be acceptable. Failure to produce a light beam parallel to within 1.0 degree to the optical axis of the receiver shall constitute failure of this test.

4.5.5 Infrared Filter. - Place the filter in a spectrophotometer and record the readings at the following wavelength:

	TRANSMITTANCE	
MICRONS	LOWER LIMIT	UPPER LIMIT
1.03	60%	None
0.9	20%	None
0.8	None	5%
0.72	None	.005%

Failure of the infrared filter transmittance to fall within the upper and lower limits shall constitute failure of the test.

4.5.6 Electromagnetic Compatibility. - The preproduction model, equipped for the reduction of electromagnetic interference in accordance with 3.10 shall be tested in accordance with MIL-E-55301, Tactical Support Class. The supplier shall furnish the contracting officer the test report and test plan required by MIL-E-55301 concurrent with the preproduction metascope. Upon approval of the report by the contracting officer and providing all other requirements of this specification are met, the preproduction model shall be used as a model for all production. Disapproval of the report shall constitute failure of this test.

4.6 Inspection Comparison. - The government may select metascopes at any time during the contract production period and subject the metascopes to the examination specified in 4.3.1 and to the tests specified to reveal any manufacturing deficiencies. The tests will be performed by the Government at a site selected and approved by the Government. Metascopes selected for inspection comparison will not include the previously selected and inspected preproduction metascope.

4.6.1 Inspection Failure. - Failure of any metascope when examined and tested as specified in 4.6 may be considered cause for refusal by

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the Government to continue acceptance of metascopes until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.7 Inspection of Preparation for Delivery. - The preservation, packaging, packing, and marking shall be examined and tested to determine compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. - Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. - Each metascope, repair parts, and technical publications contained in the carrying case shall be preserved in accordance with MIL-P-116, method IIb. The carrying case shall be in the inner container. The outer container shall be a box conforming to PPP-B-636, class 2, W5c, style optional.

5.1.2 Level B. - Each metascope, repair parts, and technical publications contained in the carrying case shall be preserved in accordance with MIL-P-116, method IC-5.

5.1.3 Level C. - Each complete metascope, repair parts, and technical publications contained in the carrying case shall be preserved and packaged to afford adequate protection against deterioration and damage.

5.2 Packing. - Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. - The metascope shall be packed in close-fitting boxes conforming to PPP-B-636, class 2, W3c, style optional. The gross weight of each box shall not exceed the weight limitations of the box specification. After closure, the box shall be waterproof-sealed with tape and the box reinforced with bands of reinforced tape in accordance with the appendix to the box specification.

5.2.2 Level B. - The metascope shall be packed in boxes conforming to PPP-B-636, type I, class 2, style optional.

5.2.3 Level C. - The metascope shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in containers complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

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5.3 Marking. - Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. - The metascope is intended for use primarily as a countermeasure aid in the night detection of near-infrared light-sources and secondarily as a general-purpose infrared-viewer.

6.2 Ordering Data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When the contractor will not furnish the batteries (see 3.9).
(c) Technical publications to be furnished (see 3.13).
(d) Repair parts to be furnished (see 3.14).
(e) Level of preservation and packaging and level of packing required (see 5.1 and 5.2). Level B preservation and packaging is intended to provide limited but economical protection and should be specified only when it is determined the metascope will be held in covered storage approximately 1 year from date of initial preservation and packaging.

6.3 Description of Test Equipment

6.3.1 The photometer should be the same or equal to the following:

Illumination meter and probe viscor-filtered	Weston Electrical Instrument Corp., Model 756
Densichron (with linear scale) and blue probe (S4) (minus aperture attachment)	W. M. Welch Scientific Co.

6.4 Definitions. - The following definitions shall apply throughout this specification.

6.4.1 Temperature Tolerance. - All temperatures in this specification assume tolerance of plus or minus 5°F unless stated otherwise in the text.

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6.4.2 Damage. - Breakage, loosening, failure, deterioration, or corrosion of any finish, hardware, connection or component; condensation of water within the receiver or infrared light-source; or failure to operate in accordance with the provisions of this specification.

CUSTODIAN:

ARMY - EL

REVIEW ACTIVITY:

ARMY - WC, MU

USER ACTIVITY:

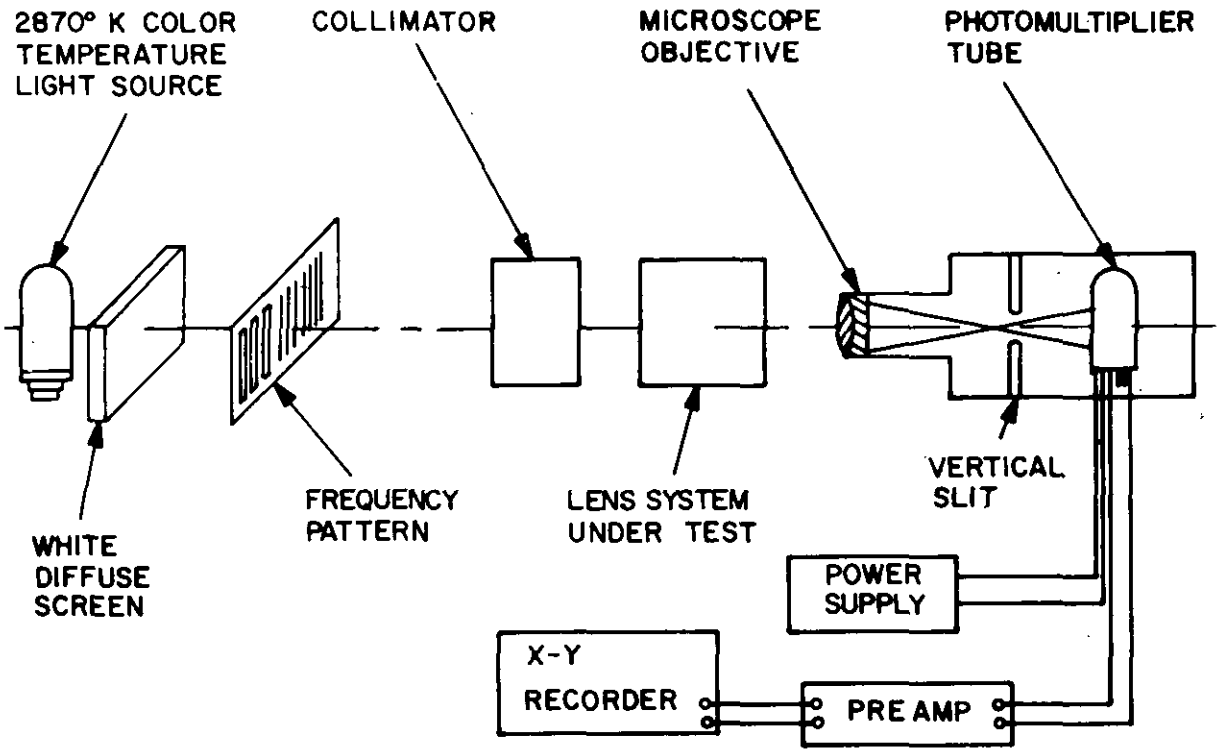
ARMY - MD

PREPARING ACTIVITY

ARMY - EL

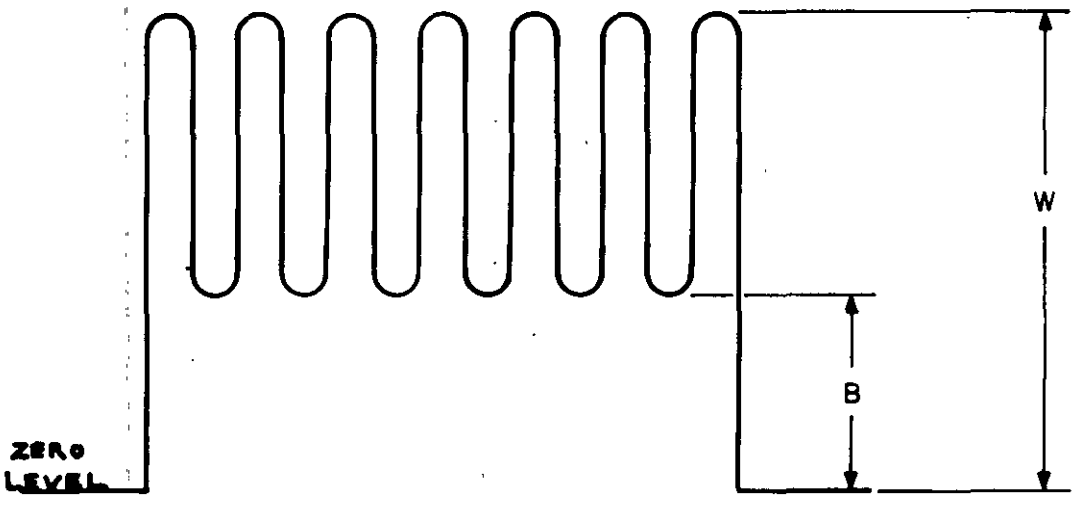
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CONTRAST TRANSFER TEST SYSTEM
FIGURE 1

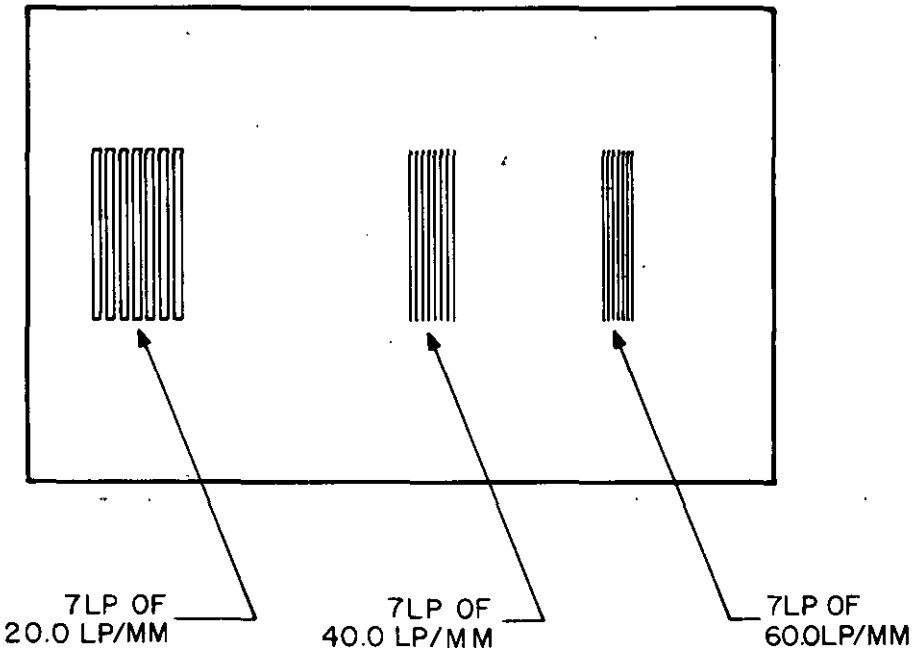
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NOTE: CONTRAST TRANSFER IS DEFINED AS $\frac{W-B}{W+B} \times 100$

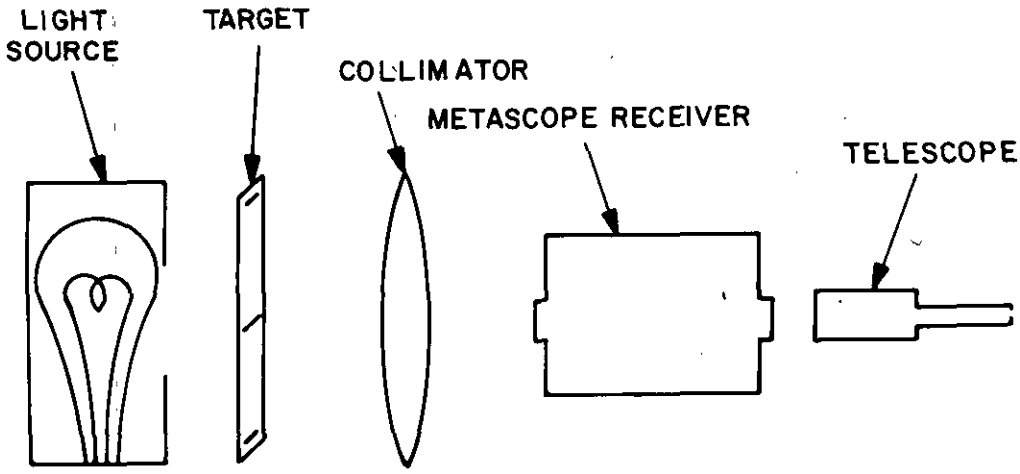
CONTRAST TRANSFER WAVE FORM
FIGURE 2

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FREQUENCY PATTERN FOR CONTRAST TRANSFER TEST
FIGURE 3

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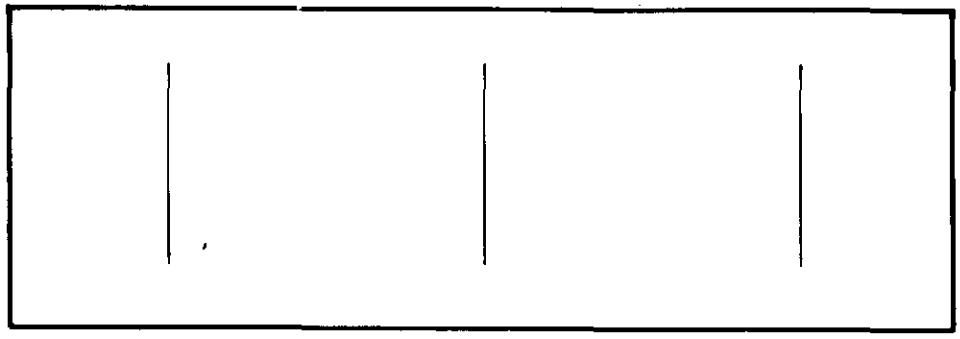
NOTES:

1. THE LIGHT SOURCE SHALL BE AS SPECIFIED IN FIGURE 1.
2. THE TARGET SHALL BE AS SHOWN IN FIGURE 5.
3. THE TELESCOPE SHALL BE NOT MORE THAN TWO (2) POWER WITH AN ENTRANCE PUPIL OF 7 MILLIMETERS AND CROSS HAIRS IN THE EYE PIECE.
4. THE TELESCOPE SHALL ALSO HAVE PROVISIONS FOR MEASURING ANGLES.

LINEAR DISTORTION TEST SYSTEM

FIGURE 4

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NOTES:

1. ALL LINES 3 MILLIMETERS IN HEIGHT.
2. THE WIDTH OF LINES IS NOT CRITICAL.
3. THE LINES TO RIGHT AND LEFT OF CENTER ARE EACH 12 DEGREES FROM CENTER LINE.

DISTORTION TEST TARGET
FIGURE 5

*U.S. GOVERNMENT PRINTING OFFICE: 1969-343-227/1321

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-45531B(EL) Metascope AN/PAS-6

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 56 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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